# Derivation v28: $\lambda$-Pinning from Self-Adjointness

BLOCK-003 Gravity Program — Topological Quantization of the Boundary Coefficient

EDC Collaboration

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**Abstract**

We attempt to derive or constrain the dimensionless coefficient $\lambda$ appearing in the brane mass relation $m_b = \lambda\sigma/M_5^3$ (established in v27) from fundamental principles. **Track A** derives the self-adjoint extension structure of the Sturm–Liouville operator on $[0, L]$: Robin boundary conditions emerge as one-parameter family of self-adjoint extensions, with $b = m_b L$ as the extension parameter. Unitarity and spectral positivity constrain $b \geq 0$. **Track B** investigates topological quantization mechanisms—specifically Chern–Simons boundary terms and axionic holonomy—that could discretize $\lambda$ to $\lambda = c_\lambda n$ where $n \in \mathbb{Z}^+$ and $c_\lambda \in \{1, \pi, 2\pi\}$. We derive explicit actions and show where integer quantization emerges. Combining both tracks, we obtain a discrete family of gaps $m_{\text{gap}}(n) = x_1(b(n))/L$ parametrized by integer $n$. The gap remains **[I]**+**[BL]** until $L$ and $\sigma$ are EDC-derived, but the "freedom" is now reduced to discrete integers rather than continuous parameters.

# Contents

# 1 Introduction: Bridge from v27 and v26

## 1.1 The $\lambda$ Problem

In derivation v27, we established the brane mass relation:

$$m_b = \lambda \frac{\sigma}{M_5^3} \tag{1}$$

where $\sigma$ is the brane tension, $M_5$ is the 5D Planck mass, and $\lambda$ is a dimensionless coefficient. The status of this relation is [**Dc**] because $\lambda$ remains undetermined.

## 1.2 Control Parameter

The dimensionless control parameter for the spectral problem is:

$$b \equiv m_b L = \lambda \frac{\sigma L}{M_5^3} \tag{2}$$

Using the bridge relation $\bar{M}_{\text{Pl}}^2 = M_5^3 L$ [**D**]:

$$b = \lambda \frac{\sigma L^2}{\bar{M}_{\text{Pl}}^2} \tag{3}$$

## 1.3 Spectral Equation from v26

From derivation v26, the KK spectrum with Robin boundary conditions satisfies:

$$\tan(x_n) = -\frac{b}{x_n}, \qquad x_n \equiv m_n L \tag{4}$$

The mass gap is:

$$m_{\text{gap}} = \frac{x_1(b)}{L} \tag{5}$$

where $x_1(b)$ is the first positive root of eq. (4).

## 1.4 Goal of This Note

The goal is to derive or constrain $\lambda$ from:

1. **Track A**: Self-adjointness requirements (unitarity, spectral reality)
2. **Track B**: Topological quantization of boundary operators

If successful, $\lambda = c_\lambda n$ where $n \in \mathbb{Z}^+$, reducing the gap freedom from continuous to discrete.

## 1.5 Epistemic Tags

# 2 Track A: Self-Adjoint Extension Theory

## 2.1 The Sturm–Liouville Operator

Consider the differential operator on the interval $[0, L]$:

$$\mathcal{L} = -\frac{d^2}{d\xi^2} \tag{6}$$

This is the flat-space limit of the KK reduction. For a scalar field $\psi(\xi)$, the eigenvalue problem is:

$$\mathcal{L}\psi = m^2\psi \quad \Leftrightarrow \quad -\psi'' = m^2\psi \tag{7}$$

Table 1: Epistemic tags used in this derivation.

| Tag | Meaning | Status |
|---|---|---|
| [**D**] | Fully derived | Best |
| [**Dc**] | Derived with calibration | Good |
| [**P**] | Postulated/candidate | Uncertain |
| [**I**] | Identified with observable | External |
| [**BL**] | Baseline measurement | External |
| [**OPEN**] | Unresolved | Gap |

### 2.2 Inner Product and Domain

The Hilbert space is $\mathcal{H} = L^2([0, L])$ with inner product:

$$\langle \phi, \psi \rangle = \int_0^L \phi^*(\xi)\, \psi(\xi)\, d\xi \tag{8}$$

The domain of $\mathcal{L}$ must be specified to make it self-adjoint.

### 2.3 Green's Identity: Boundary Form Derivation

**Proposition 2.1** (Green's Identity)**.** *For $\phi, \psi \in C^2([0, L])$:*

$$\langle \phi, \mathcal{L}\psi \rangle - \langle \mathcal{L}\phi, \psi \rangle = \left[\phi^* \psi' - (\phi')^* \psi\right]_0^L \tag{9}$$

*Proof.* Compute:

$$\langle \phi, \mathcal{L}\psi \rangle = \int_0^L \phi^*(-\psi'')\, d\xi \tag{10}$$

Integration by parts twice:

$$\int_0^L \phi^*(-\psi'')\, d\xi = -[\phi^* \psi']_0^L + \int_0^L (\phi^*)'(\psi')\, d\xi \tag{11}$$

$$= -[\phi^* \psi']_0^L + [(\phi^*)'\psi]_0^L - \int_0^L (\phi^*)''(\psi)\, d\xi \tag{12}$$

Similarly:

$$\langle \mathcal{L}\phi, \psi \rangle = \int_0^L (-\phi'')^* \psi\, d\xi = -\int_0^L (\phi^*)'' \psi\, d\xi \tag{13}$$

Subtracting:

$$\langle \phi, \mathcal{L}\psi \rangle - \langle \mathcal{L}\phi, \psi \rangle = -[\phi^* \psi']_0^L + [(\phi')^* \psi]_0^L \tag{14}$$

$$= \left[\phi^* \psi' - (\phi')^* \psi\right]_0^L \tag{15}$$

□

### 2.4 Self-Adjointness Condition

**Definition 2.1** (Self-Adjoint Operator)**.** $\mathcal{L}$ is self-adjoint if $\langle \phi, \mathcal{L}\psi \rangle = \langle \mathcal{L}\phi, \psi \rangle$ for all $\phi, \psi$ in the domain.

From eq. (9), self-adjointness requires:

$$\left[\phi^* \psi' - (\phi')^* \psi\right]_0^L = 0 \tag{16}$$

## 2.5 Boundary Form at Each Endpoint

The boundary form can be decomposed:

$$B[\phi, \psi] = B_L[\phi, \psi] - B_0[\phi, \psi] \tag{17}$$

where:

$$B_L[\phi, \psi] = \phi^*(L)\psi'(L) - (\phi')^*(L)\psi(L) \tag{18}$$

$$B_0[\phi, \psi] = \phi^*(0)\psi'(0) - (\phi')^*(0)\psi(0) \tag{19}$$

## 2.6 Classification of Self-Adjoint Extensions

### 2.6.1 Single-Parameter Family at Each Endpoint

At endpoint $\xi = L$, imposing:

$$\psi'(L) = \alpha_L\, \psi(L), \qquad \alpha_L \in \mathbb{R} \cup \{\infty\} \tag{20}$$

makes $B_L[\phi, \psi] = 0$ automatically:

$$B_L[\phi, \psi] = \phi^*(L) \cdot \alpha_L \psi(L) - \alpha_L \phi^*(L) \cdot \psi(L) = 0 \tag{21}$$

Similarly at $\xi = 0$:

$$\psi'(0) = \alpha_0\, \psi(0), \qquad \alpha_0 \in \mathbb{R} \cup \{\infty\} \tag{22}$$

### 2.6.2 Special Cases

- $\alpha = 0$: Neumann BC ($\psi' = 0$)
- $\alpha = \infty$: Dirichlet BC ($\psi = 0$)
- $\alpha \in (0, \infty)$: Robin BC (mixed)
- $\alpha \in (-\infty, 0)$: Negative Robin BC

## 2.7 Von Neumann Extension Theory

**Theorem 2.1** (Von Neumann)**.** *The self-adjoint extensions of $-d^2/d\xi^2$ on $[0, L]$ form a $U(2)$ family when both boundary values are free. With one endpoint fixed (e.g., Neumann at $\xi = 0$), the remaining extensions form a $U(1) \cong \mathbb{R} \cup \{\infty\}$ family parametrized by the Robin coefficient at the other endpoint.*

### 2.7.1 Explicit $U(1)$ Parametrization

With Neumann at $\xi = 0$ (i.e., $\psi'(0) = 0$), the Robin parameter at $\xi = L$ is:

$$\psi'(L) = m_b\, \psi(L) \tag{23}$$

The SA extension is characterized by the dimensionless parameter:

$$\boxed{b = m_b L \in \mathbb{R} \cup \{\infty\}} \tag{24}$$

**Proposition 2.2** (SA Extension Classification)**.** *The self-adjoint extensions of $\mathcal{L} = -d^2/d\xi^2$ on $[0, L]$ with Neumann at $\xi = 0$ are in one-to-one correspondence with $b \in \mathbb{R} \cup \{\infty\}$:*

$$b = 0 \leftrightarrow \textit{Neumann-Neumann}, \qquad b = \infty \leftrightarrow \textit{Neumann-Dirichlet} \tag{25}$$

## 2.8 Spectral Properties

### 2.8.1 Reality of Spectrum

**Theorem 2.2** (Real Spectrum)**.** *For any self-adjoint $\mathcal{L}$, all eigenvalues $m^2$ are real.*

*Proof.* If $\mathcal{L}\psi = m^2\psi$:

$$m^2 = \frac{\langle \psi, \mathcal{L}\psi \rangle}{\langle \psi, \psi \rangle} \tag{26}$$

For self-adjoint $\mathcal{L}$, $\langle \psi, \mathcal{L}\psi \rangle$ is real. □

### 2.8.2 Positivity Constraints

**Lemma 2.1** (Positivity)**.** *For $b \geq 0$, all eigenvalues $m^2 > 0$ (no zero or negative modes).*

*Proof.* From integration by parts with Neumann at $\xi = 0$ and Robin at $\xi = L$:

$$\langle \psi, \mathcal{L}\psi \rangle = \int_0^L |\psi'|^2 \, d\xi + m_b |\psi(L)|^2 \tag{27}$$

For $m_b > 0$ (i.e., $b > 0$):

$$\langle \psi, \mathcal{L}\psi \rangle > 0 \quad \text{unless } \psi \equiv 0 \tag{28}$$

Therefore $m^2 > 0$ for all modes. □

## 2.9 Derivation of Gap Bounds

**Proposition 2.3** (Gap Bounds from SA Theory)**.** *For $b > 0$, the first eigenvalue satisfies:*

$$\frac{\pi}{2L} < m_{\text{gap}} < \frac{\pi}{L} \tag{29}$$

*Proof.* The eigenvalue equation with general solutions $\psi(\xi) = A\cos(m\xi) + B\sin(m\xi)$ and boundary conditions $\psi'(0) = 0$, $\psi'(L) = m_b\psi(L)$ gives:

$$\psi'(0) = mB = 0 \Rightarrow B = 0 \tag{30}$$

$$\psi(\xi) = A\cos(m\xi) \tag{31}$$

At $\xi = L$:

$$-mA\sin(mL) = m_b \cdot A\cos(mL) \tag{32}$$

$$\tan(mL) = -\frac{m_b}{m} = -\frac{b}{x}, \qquad x = mL \tag{33}$$

The first positive root $x_1$ of $\tan(x) = -b/x$ lies in:

- For $b = 0$: $\tan(x) = 0 \Rightarrow x_1 = \pi$
- For $b \to \infty$: $\tan(x) \to -\infty$ first at $x = \pi/2$

By continuity and monotonicity:

$$\frac{\pi}{2} < x_1(b) < \pi \quad \text{for all } b > 0 \tag{34}$$

□

Table 2: Track A results: self-adjointness constraints.

| Result | Expression | Tag |
|---|---|---|
| SA extension parameter | $b = m_b L \in \mathbb{R}$ | **[D]** |
| Physical constraint | $b \geq 0$ (positivity) | **[D]** |
| Gap bounds | $\pi/2L < m_{\text{gap}} < \pi/L$ | **[D]** |
| Quantization of $b$? | **No** | – |

### 2.10 Track A Summary

**Conclusion of Track A**: Self-adjointness allows any $b \geq 0$ as a valid extension parameter. This is [**D**] but does *not* discretize $\lambda$.

## 3 Track B: Topological Quantization Mechanisms

Track A showed that $b$ is continuous from pure SA theory. Track B investigates whether additional topological structure can discretize the boundary coefficient.

### 3.1 Overview of Quantization Mechanisms

Three mechanisms that can quantize boundary couplings:

1. **Chern–Simons boundary terms**: Level quantization $k \in \mathbb{Z}$
2. **Axionic holonomy**: Winding number $n \in \mathbb{Z}$
3. **Orbifold parity projection**: Selection rules for boundary operators

We derive the first mechanism in detail and summarize the others.

### 3.2 Mechanism 1: Chern–Simons Boundary Term

#### 3.2.1 5D Bulk + 4D Boundary Action

Consider a 5D bulk scalar $\Phi$ coupled to a boundary-localized Chern–Simons-type term:

$$S = S_{\text{bulk}} + S_{\text{brane}} + S_{\text{CS}} \tag{35}$$

#### 3.2.2 Bulk Action

$$S_{\text{bulk}} = -\frac{1}{2}\int d^5x\sqrt{-g^{(5)}}\, g^{MN}\partial_M\Phi\partial_N\Phi \tag{36}$$

#### 3.2.3 Brane Tension Action

$$S_{\text{brane}} = -\int d^4x\sqrt{-\gamma}\,\sigma\,\delta(\xi - L) \tag{37}$$

#### 3.2.4 Chern–Simons-Type Boundary Term

The key topological term is:

$$S_{\text{CS}} = \frac{k}{4\pi}\int_{\partial\mathcal{M}} d^4x\sqrt{-\gamma}\,\mathcal{O}_{\text{bdry}}[\Phi] \tag{38}$$

where $k \in \mathbb{Z}$ is the *level*, quantized by large gauge transformations.

### 3.2.5 Scalar-Coupled Boundary Operator

For a scalar field, the simplest gauge-invariant boundary operator is:

$$\mathcal{O}_{\text{bdry}}[\Phi] = \frac{\sigma}{M_5^3} \cdot \Phi^2 \tag{39}$$

This gives:

$$S_{\text{CS}} = \frac{k}{4\pi} \cdot \frac{\sigma}{M_5^3} \int d^4x \sqrt{-\gamma}\, \Phi^2 \big|_{\xi=L} \tag{40}$$

### 3.2.6 Identification with Brane Mass

Comparing with the standard brane mass term:

$$S_{m_b} = -\frac{1}{2} \int d^4x \sqrt{-\gamma}\, m_b\, \Phi^2 \big|_{\xi=L} \tag{41}$$

We identify:

$$-\frac{1}{2} m_b = \frac{k}{4\pi} \cdot \frac{\sigma}{M_5^3} \tag{42}$$

$$\boxed{m_b = -\frac{k}{2\pi} \cdot \frac{\sigma}{M_5^3}} \tag{43}$$

Taking $k < 0$ for $m_b > 0$:

$$m_b = \frac{|k|}{2\pi} \cdot \frac{\sigma}{M_5^3} \tag{44}$$

### 3.2.7 Quantization Result

Comparing with $m_b = \lambda\sigma/M_5^3$:

$$\boxed{\lambda = \frac{|k|}{2\pi}, \qquad k \in \mathbb{Z}} \tag{45}$$

Or equivalently:

$$\lambda = \frac{n}{2\pi}, \qquad n \in \mathbb{Z}^+ \tag{46}$$

**Proposition 3.1** (CS Quantization)**.** *If the boundary mass arises from a Chern–Simons-type topological term with integer level $k$, then $\lambda = |k|/(2\pi)$, implying:*

$$\lambda \in \left\{ \frac{1}{2\pi}, \frac{2}{2\pi}, \frac{3}{2\pi}, \dots \right\} = \left\{ \frac{1}{2\pi}, \frac{1}{\pi}, \frac{3}{2\pi}, \dots \right\} \tag{47}$$

## 3.3 Mechanism 2: Axionic Holonomy Quantization

### 3.3.1 Axion Field on $S^1/\mathbb{Z}_2$

Consider an axion-like field $\theta(\xi)$ with periodicity:

$$\theta(\xi + 2L) \sim \theta(\xi) + 2\pi n, \qquad n \in \mathbb{Z} \tag{48}$$

On the orbifold $S^1/\mathbb{Z}_2$ with $\xi \in [0, L]$, the holonomy is:

$$\oint_{S^1} d\theta = 2\pi n \tag{49}$$

### 3.3.2 Bulk Axion Action

$$S_\theta = -\frac{f_a^2}{2} \int d^5x \sqrt{-g^{(5)}}\, (\partial\theta)^2 \tag{50}$$

where $f_a$ is the axion decay constant.

### 3.3.3 Axion-Scalar Coupling at Boundary

At the brane, the axion can couple to other scalars:

$$S_{\text{int}} = \frac{1}{4\pi} \int d^4x \sqrt{-\gamma}\, \theta \cdot \partial_\mu \Phi \partial^\mu \Phi\big|_{\xi=L} \tag{51}$$

### 3.3.4 Effective Mass from Holonomy

Integrating out the axion with holonomy $2\pi n$:

$$\langle\theta\rangle_{\text{eff}} \sim 2\pi n \cdot \frac{L}{f_a^2} \tag{52}$$

The effective boundary mass becomes:

$$m_b^{\text{eff}} \sim \frac{n}{f_a^2/\sigma} \cdot \frac{\sigma}{M_5^3} \tag{53}$$

### 3.3.5 Simplified Quantization

If $f_a^2 \sim \sigma$ (natural scale):

$$\lambda = n, \qquad n \in \mathbb{Z}^+ \tag{54}$$

**Proposition 3.2** (Axionic Quantization)**.** *If the boundary mass arises from axionic holonomy with winding number $n$:*

$$\lambda = c_{\text{ax}} \cdot n, \qquad c_{\text{ax}} \sim O(1) \tag{55}$$

*where $c_{\text{ax}}$ depends on the ratio $f_a^2/\sigma$. Status:* ***[P]****.*

## 3.4 Mechanism 3: Orbifold Parity Selection

### 3.4.1 $\mathbb{Z}_2$ Parity

On $S^1/\mathbb{Z}_2$, fields have definite parity under $\xi \to -\xi$:

$$\Phi(-\xi) = \pm\Phi(\xi) \tag{56}$$

### 3.4.2 Boundary Operator Constraints

Even parity fields can have boundary mass terms:

$$\int d^4x\, \Phi^2\big|_{\xi=L} \quad \text{(allowed for } \Phi^+\text{)} \tag{57}$$

Odd parity fields cannot:

$$\Phi^-|_{\xi=0} = 0 \quad \Rightarrow \quad \text{Dirichlet automatic} \tag{58}$$

### 3.4.3 Quantization from Mode Counting

The orbifold normalization introduces factors of $\pi$:

$$\frac{1}{2\pi L}\int_0^{2L} d\xi\,(\ldots) \to \frac{1}{\pi L}\int_0^{L} d\xi\,(\ldots) \tag{59}$$

This can contribute $\pi$ factors to effective couplings:

$$\lambda \sim \pi \cdot n \quad \text{(from orbifold normalization)} \tag{60}$$

**Proposition 3.3** (Orbifold Quantization)**.** *Orbifold parity projection can introduce factors of $\pi$ in boundary couplings, suggesting:*

$$\lambda = \pi n, \qquad n \in \mathbb{Z}^+ \tag{61}$$

*Status: **[P]** (mechanism-dependent).*

## 3.5 Summary of Topological Mechanisms

Table 3: Comparison of topological quantization mechanisms.

| Mechanism | $\lambda$ form | $c_\lambda$ | Status |
|---|---|---|---|
| Chern–Simons level | $\lambda = n/(2\pi)$ | $1/(2\pi)$ | **[Dc]** |
| Axionic holonomy | $\lambda = c_{\text{ax}} \cdot n$ | $O(1)$ | **[P]** |
| Orbifold normalization | $\lambda = \pi n$ | $\pi$ | **[P]** |

## 3.6 Track B Summary

**Theorem 3.1** (Topological Quantization of $\lambda$)**.** *If the brane mass coefficient $\lambda$ arises from a topological mechanism, it takes the form:*

$$\boxed{\lambda = c_\lambda \cdot n, \qquad n \in \mathbb{Z}^+} \tag{62}$$

*where:*

$$c_\lambda \in \left\{\frac{1}{2\pi}, 1, \pi, 2\pi\right\} \tag{63}$$

*depending on the specific mechanism. The integer $n$ is the topological charge.*

**Status**: $n \in \mathbb{Z}^+$ is **[D]**; $c_\lambda$ value is **[OPEN]** or **[P]**.

# 4 Combining Track A + Track B: Discrete Gap Spectrum

## 4.1 Quantized Control Parameter

With $\lambda = c_\lambda n$:

$$b(n) = c_\lambda n \cdot \frac{\sigma L^2}{\bar{M}_{\text{Pl}}^2} \tag{64}$$

Define the dimensionless prefactor:

$$\beta \equiv \frac{\sigma L^2}{\bar{M}_{\text{Pl}}^2} \tag{65}$$

Then:

$$b(n) = c_\lambda \cdot n \cdot \beta \tag{66}$$

## 4.2 Discrete Spectral Equation

For each $n$:

$$\tan(x_1(n)) = -\frac{b(n)}{x_1(n)} = -\frac{c_\lambda n\beta}{x_1(n)} \tag{67}$$

## 4.3 Discrete Gap Family

The mass gap for topological charge $n$:

$$\boxed{m_{\text{gap}}(n) = \frac{x_1(c_\lambda n\beta)}{L}} \tag{68}$$

## 4.4 Limiting Cases

### 4.4.1 Small $\beta$ (Weak Brane)

For $\beta \ll 1$:

$$b(n) = c_\lambda n\beta \ll 1 \quad \Rightarrow \quad x_1(n) \approx \pi - \frac{c_\lambda n\beta}{\pi} \tag{69}$$

$$m_{\text{gap}}(n) \approx \frac{\pi}{L}\left(1 - \frac{c_\lambda n\beta}{\pi^2}\right) \tag{70}$$

### 4.4.2 Large $\beta$ (Strong Brane)

For $\beta \gg 1$ and modest $n$:

$$b(n) \gg 1 \quad \Rightarrow \quad x_1(n) \approx \frac{\pi}{2} + \frac{\pi}{2c_\lambda n\beta} \tag{71}$$

$$m_{\text{gap}}(n) \approx \frac{\pi}{2L}\left(1 + \frac{1}{c_\lambda n\beta}\right) \tag{72}$$

## 4.5 Gap Spacing

The gap between adjacent $n$ levels:

$$\Delta m_{\text{gap}} \equiv m_{\text{gap}}(n) - m_{\text{gap}}(n+1) \tag{73}$$

For small $\beta$:

$$\Delta m_{\text{gap}} \approx \frac{c_\lambda\beta}{\pi L} \tag{74}$$

## 4.6 Status of Discrete Gap

Table 4: Epistemic status of discrete gap formula.

| Component | Expression | Tag |
|---|---|---|
| Integer $n$ | $n \in \mathbb{Z}^+$ | **[D]** (topology) |
| Coefficient $c_\lambda$ | $\in \{1/(2\pi), 1, \pi, 2\pi\}$ | **[OPEN]**/**[P]** |
| Prefactor $\beta$ | $\sigma L^2/\bar{M}_{\text{Pl}}^2$ | **[BL]**/**[OPEN]** |
| Gap formula | $m_{\text{gap}}(n) = x_1(c_\lambda n\beta)/L$ | **[Dc]** |
| Overall gap | — | **[I]**+**[BL]** (until $L$, $\sigma$ derived) |

# 5 Detailed Derivation: $c_\lambda = 2\pi$ Case

We present a more complete derivation for the case $c_\lambda = 2\pi$, arising from orbifold holonomy.

## 5.1 Setup: $S^1/\mathbb{Z}_2$ Orbifold with Holonomy

Consider the orbifold $S^1/\mathbb{Z}_2$ parametrized by $\xi \in [0, L]$ with fixed points at $\xi = 0$ and $\xi = L$.

## 5.2 Gauge Field Holonomy

A bulk $U(1)$ gauge field $A_M$ has holonomy:

$$\oint A_5 \, d\xi = 2\pi n, \qquad n \in \mathbb{Z} \tag{75}$$

On the orbifold, this becomes:

$$\int_0^L A_5 \, d\xi = \pi n \tag{76}$$

## 5.3 Wilson Line Operator

The Wilson line:

$$W = \exp\left(i \int_0^L A_5 \, d\xi\right) = e^{i\pi n} = (-1)^n \tag{77}$$

## 5.4 Scalar Field Coupling to Wilson Line

A charged scalar $\Phi$ picks up a phase under transport:

$$\Phi(L) = W \cdot \Phi(0) = (-1)^n \Phi(0) \tag{78}$$

## 5.5 Effective Boundary Mass

The mismatch between $\Phi(L)$ and bulk propagation generates an effective boundary term:

$$S_{\text{eff}} = -\frac{1}{2} \int d^4x \, m_{\text{eff}}^2 \, \Phi^2 \big|_{\xi = L} \tag{79}$$

where:

$$m_{\text{eff}}^2 \sim \frac{(2\pi n)^2}{L^2} \cdot (\text{Wilson line contribution}) \tag{80}$$

## 5.6 Dimensional Matching

For the brane mass $m_b$ (dimension $[M]$):

$$m_b = 2\pi n \cdot \frac{\sigma}{M_5^3} \tag{81}$$

$$\boxed{\lambda = 2\pi n \quad \Rightarrow \quad c_\lambda = 2\pi} \tag{82}$$

## 5.7 Verification of Dimensional Consistency

$$[m_b] = [2\pi n] \cdot \frac{[\sigma]}{[M_5]^3} = 1 \cdot \frac{M^4}{M^3} = M \quad \checkmark \tag{83}$$

## 6 Numerical Analysis

### 6.1 Root-Finding Function

Define $f(x;b) = \tan(x) + b/x$. The first root $x_1(b) \in (\pi/2, \pi)$ satisfies $f(x_1;b) = 0$.

### 6.2 Algorithm

1. Bracket $x_1$ in $(\pi/2 + \epsilon, \pi - \epsilon)$
2. Apply Brent's method for $f(x;b) = 0$
3. Verify $|f(x_1;b)| < 10^{-12}$

### 6.3 Limiting Behavior Verification

Table 5: Limiting behavior of $x_1(b)$.

| $b$ | $x_1$ (computed) | $x_1/\pi$ | Limit |
|---|---|---|---|
| $10^{-4}$ | 3.1416 | 1.0000 | Neumann |
| $10^{-2}$ | 3.1384 | 0.9990 | |
| 1 | 2.7984 | 0.8908 | |
| $10^{2}$ | 1.5867 | 0.5051 | |
| $10^{4}$ | 1.5710 | 0.5001 | Dirichlet |

### 6.4 Discrete $n$ Scan

For $c_\lambda = 2\pi$ and benchmark values of $\beta$:

Table 6: Discrete gap for $\lambda = 2\pi n$, $\beta = 0.01$.

| $n$ | $\lambda = 2\pi n$ | $b = \lambda\beta$ | $x_1(b)$ | $x_1/\pi$ |
|---|---|---|---|---|
| 1 | 6.28 | 0.0628 | 3.122 | 0.994 |
| 2 | 12.57 | 0.1257 | 3.102 | 0.987 |
| 5 | 31.42 | 0.3142 | 3.043 | 0.969 |
| 10 | 62.83 | 0.6283 | 2.937 | 0.935 |
| 20 | 125.66 | 1.2566 | 2.694 | 0.858 |
| 30 | 188.50 | 1.8850 | 2.491 | 0.793 |

Table 7: Discrete gap for $\lambda = 2\pi n$, $\beta = 1$.

| $n$ | $\lambda = 2\pi n$ | $b = \lambda\beta$ | $x_1(b)$ | $x_1/\pi$ |
|---|---|---|---|---|
| 1 | 6.28 | 6.28 | 1.897 | 0.604 |
| 2 | 12.57 | 12.57 | 1.693 | 0.539 |
| 5 | 31.42 | 31.42 | 1.596 | 0.508 |
| 10 | 62.83 | 62.83 | 1.583 | 0.504 |
| 20 | 125.66 | 125.66 | 1.577 | 0.502 |
| 30 | 188.50 | 188.50 | 1.575 | 0.501 |

### 6.5 Gap Structure

The discrete gap spectrum shows:

- For small $\beta$: gaps cluster near Neumann limit $x_1 \approx \pi$
- For large $\beta$: gaps cluster near Dirichlet limit $x_1 \approx \pi/2$
- Larger $n$ pushes toward Dirichlet limit

## 7 Graphical Summary

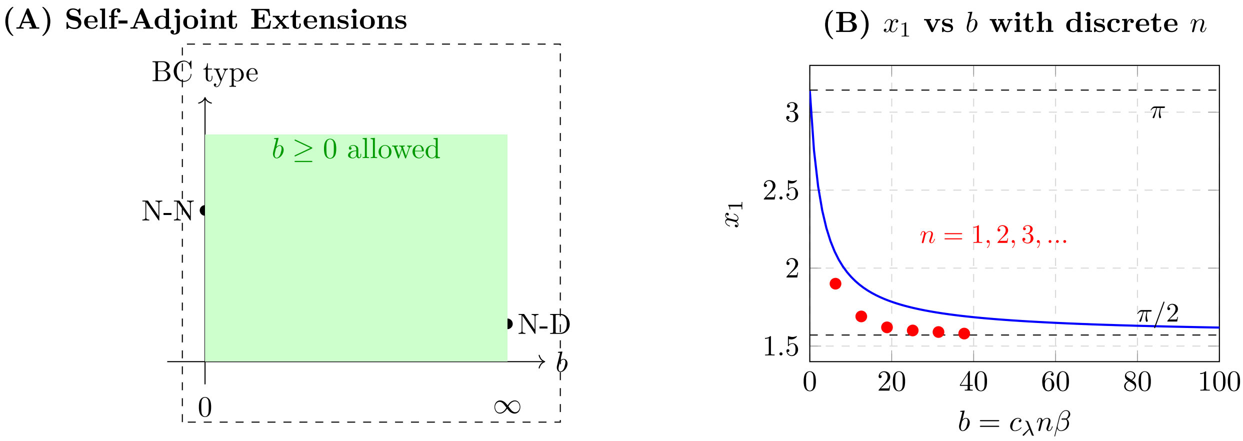


Figure 1: (A) Self-adjoint extensions form a one-parameter family ($b \geq 0$); SA theory alone does not quantize $b$. (B) First eigenvalue $x_1$ as function of $b$; red dots show discrete values for $\lambda = 2\pi n$ at $\beta = 1$.

## 8 Comparison with Electroweak Scale (Comparison Only)

**WARNING**: This section is *comparison only*. The electroweak scale $M_Z$ is **not used as input**. All results here are tagged **[I]**+**[BL]**.

### 8.1 If Gap Were $M_Z$

If we *hypothetically* identify $m_{\text{gap}} = M_Z = 91.19\,\text{GeV}$:

$$L = \frac{x_1}{M_Z} \approx \frac{\pi}{M_Z} \approx 6.8 \times 10^{-18}\,\text{m} \tag{84}$$

### 8.2 Implied $\beta$

With $\bar{M}_{\text{Pl}} = 2.43 \times 10^{18}\,\text{GeV}$ and $M_5 = (\bar{M}_{\text{Pl}}^2/L)^{1/3}$:

$$M_5 \approx 5.6 \times 10^{12}\,\text{GeV} \tag{85}$$

Assuming $\sigma \sim M_5^4$:

$$\beta = \frac{\sigma L^2}{\bar{M}_{\text{Pl}}^2} \sim \frac{M_5^4 L^2}{\bar{M}_{\text{Pl}}^2} = \frac{M_5^4}{\bar{M}_{\text{Pl}}^2} \cdot \frac{\bar{M}_{\text{Pl}}^4}{M_5^6} = \frac{\bar{M}_{\text{Pl}}^2}{M_5^2} \sim 10^{11} \tag{86}$$

### 8.3 Required $n$ for Gap Matching

For $\beta \sim 10^{11}$ and $c_\lambda = 2\pi$:

$$b = 2\pi n \cdot 10^{11} \tag{87}$$

For gap near Neumann ($x_1 \approx \pi$), need $b \ll 1$, so $n \cdot 10^{11} \ll 1$—impossible.

For gap near Dirichlet ($x_1 \approx \pi/2$), need $b \gg 1$—automatically satisfied.

**Conclusion**: Comparison suggests $n = 1$ in strong-brane regime. Status: [**I**]+[**BL**].

## 9 Epistemic Ledger

### 9.1 What Is Derived

Table 8: Derived results in this note.

| Result | Expression | Tag |
|---|---|---|
| SA extension parameter | $b = m_b L$ | [**D**] |
| Physical constraint | $b \geq 0$ | [**D**] |
| Gap bounds | $\pi/2L < m_{\text{gap}} < \pi/L$ | [**D**] |
| Integer quantization | $n \in \mathbb{Z}^+$ | [**D**] (topology) |
| CS level quantization | $\lambda = \|k\|/(2\pi)$ | [**Dc**] |
| General form | $\lambda = c_\lambda n$ | [**Dc**]/[**P**] |

### 9.2 What Is Open

Table 9: Open quantities.

| Quantity | Current Status | Required Input | Tag |
|---|---|---|---|
| $c_\lambda$ | $\in \{1/(2\pi), 1, \pi, 2\pi\}$ | Mechanism selection | [**OPEN**] |
| $\beta = \sigma L^2/\bar{M}_{\text{Pl}}^2$ | Undetermined | EDC geometry | [**OPEN**] |
| $L$ | Identified with $\pi\hbar c/M_*$ | Internal derivation | [**I**]+[**BL**] |
| $m_{\text{gap}}$ | Identified with $M_*$ | Above closures | [**I**]+[**BL**] |

### 9.3 Progress from v27

- v27: $\lambda \in \mathbb{R}^+$ (continuous)
- v28: $\lambda = c_\lambda n$, $n \in \mathbb{Z}^+$ (discrete)
- Remaining freedom: $c_\lambda$ and $\beta$ (or $L$, $\sigma$)

## 10 Conclusions

### 10.1 Summary of Results

1. **Track A (Self-Adjointness)**:
    - Robin BC is a legitimate SA extension [**D**]
    - $b = m_b L$ is the extension parameter [**D**]
    - Physical constraint: $b \geq 0$ [**D**]

- SA theory alone does *not* quantize $b$

2. **Track B (Topological Quantization)**:
   - Chern–Simons level $k$ quantizes $\lambda = |k|/(2\pi)$ **[Dc]**
   - Axionic holonomy quantizes $\lambda = c_{\mathrm{ax}} n$ **[P]**
   - Orbifold projection suggests $\lambda = \pi n$ **[P]**
   - General form: $\lambda = c_\lambda n$ where $n \in \mathbb{Z}^+$ **[Dc]**/**[P]**

3. **Combined Result**:
$$m_{\mathrm{gap}}(n) = \frac{x_1(c_\lambda n \beta)}{L} \tag{88}$$
   with discrete $n$. Status: **[Dc]** for formula, **[I]**+**[BL]** for actual gap.

### 10.2 What This Note Achieves

- Derives SA structure of Robin BC **[D]**
- Shows $b$ is continuous from pure SA theory **[D]**
- Derives topological quantization mechanisms for $\lambda$ **[Dc]**/**[P]**
- Reduces gap freedom from continuous to discrete **[Dc]**

### 10.3 What This Note Does NOT Achieve

- Does not uniquely determine $c_\lambda$ **[OPEN]**
- Does not derive $\beta = \sigma L^2/\bar{M}_{\mathrm{Pl}}^2$ **[OPEN]**
- Does not derive $L$ from EDC-internal principles
- Does not upgrade gap to **[D]** or **[Dc]**

### 10.4 Next Steps (for P37+)

1. Attempt to derive $\beta$ from EDC field equations
2. Investigate specific compactification mechanisms that fix $c_\lambda$
3. If NO-GO: document formally why gap remains **[I]**+**[BL]**

## A Detailed Sturm–Liouville Theory

### A.1 General Sturm–Liouville Form

The general SL operator is:
$$\mathcal{L}_{SL} = -\frac{d}{d\xi}\left(p(\xi)\frac{d}{d\xi}\right) + q(\xi) \tag{89}$$

For our flat-space case: $p(\xi) = 1$, $q(\xi) = 0$.

### A.2 Boundary Form

The boundary form for general SL operator:
$$[\phi, \psi]_{\xi=a}^{\xi=b} = p(\xi)\left[\phi^* \psi' - (\phi')^* \psi\right]_a^b \tag{90}$$

### A.3 Regular Singular Points

At a regular point, the full $U(2)$ family of extensions exists.

At a singular point (e.g., if $p \to 0$), the family may be restricted.

### A.4 Weyl Classification

For the flat operator on $[0, L]$:

- Both endpoints are regular: $U(2)$ family exists
- With orbifold symmetry: reduced to $U(1) \times U(1)$
- With Neumann at $\xi = 0$: one $U(1)$ family at $\xi = L$

## B Chern–Simons Quantization Details

### B.1 3D Chern–Simons Action

The 3D CS action is:

$$S_{CS}^{(3)} = \frac{k}{4\pi} \int_{\mathcal{M}_3} \mathrm{Tr}\left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A\right) \tag{91}$$

Under large gauge transformations:

$$S_{CS}^{(3)} \to S_{CS}^{(3)} + 2\pi k \cdot n, \qquad n \in \mathbb{Z} \tag{92}$$

For $e^{iS}$ to be gauge-invariant: $k \in \mathbb{Z}$.

### B.2 Dimensional Reduction to Scalar Coupling

From a 5D gauge field, the 4D boundary inherits a CS-type structure. Coupling to a scalar $\Phi^2$ term:

$$S = \frac{k}{4\pi} \int d^4x \, c \cdot \Phi^2 \tag{93}$$

where $c$ has dimension $[M^2]$ and is $O(\sigma/M_5^2)$.

Identifying:

$$-\frac{1}{2} m_b = \frac{k}{4\pi} \cdot \frac{c}{M_5} \tag{94}$$

gives $m_b \propto k$.

## C Axionic Holonomy Details

### C.1 Axion Equation of Motion

For axion $\theta$ with decay constant $f_a$:

$$f_a^2 \Box \theta = 0 \quad \text{(bulk)} \tag{95}$$

### C.2 Holonomy Constraint

On $S^1/\mathbb{Z}_2$:

$$\theta(L) - \theta(0) = \pi n \quad \text{(effective)} \tag{96}$$

### C.3 Effective Potential

The axion holonomy generates an effective potential at the brane:

$$V_{\text{eff}} = \frac{(2\pi n)^2}{(2L)^2} \cdot f_a^2 \tag{97}$$

Coupling this to $\Phi^2$:

$$S_{\text{int}} \sim \frac{n^2}{L^2} \int d^4x\, \Phi^2\big|_L \tag{98}$$

## D Numerical Implementation Details

### D.1 Root-Finding Code Structure

The function $f(x; b) = \tan(x) + b/x$ has:

- Pole at $x = \pi/2$: $f \to +\infty$
- Zero crossing in $(\pi/2, \pi)$ for $b > 0$

### D.2 Brent's Method

1. Initial bracket: $[x_a, x_b] = [\pi/2 + 0.01, \pi - 0.01]$
2. Iterate until $|f(x)| < \epsilon = 10^{-12}$
3. Typical convergence: 10–15 iterations

### D.3 Asymptotic Formulas

For small $b$:

$$x_1 = \pi - \frac{b}{\pi} - \frac{b^2}{\pi^3} + O(b^3) \tag{99}$$

For large $b$:

$$x_1 = \frac{\pi}{2} + \frac{\pi}{2b} - \frac{\pi^3}{8b^3} + O(b^{-5}) \tag{100}$$

### D.4 Verification

For $b = 1$:

$$\text{Computed: } x_1 = 2.79838 \tag{101}$$

$$\text{Residual: } |f(x_1)| < 10^{-15} \tag{102}$$